

James Squires

414-335-9522 | James.Squires@uga.edu

EDUCATION

The University of Georgia, Morehead Honors College, Athens, GA December 2026
Bachelor of Science: Physics, Mathematics **GPA: 3.89/4.00**

RESEARCH AND DIRECTED READINGS

Categorical Quantum Deliberative Machines Athens, GA
(UGA) Research Assistant, Advisor(s): Andrei Galiautdinov Oct 2025 – Present

- Modeling Quantum Deliberation Machines (QDMs) in terms of Abramsky-Coecke semantics
- Developing exposition on Categorical Quantum Mechanics and QDMS

Statistical Learning Model in STEM Education Predictive Analytics Athens, GA
(UGA), Research Assistant, Advisor(s): Nicholas Young, Nandana Weliweriya Dec 2023 – Present

- Built logistic regression and random forest models to predict at-risk physics students
- Investigating classifying student classroom profiles via persistent homology TDA

Simulating MIR Spectra Emission of Water in Accretion Disks Athens, GA
(UGA), Research Assistant, Advisor(s): Phillip Stancil Aug 2024 – Present

- Simulated spectroscopic emissions of astrophysical objects (L1527, PDS70) using Radex
- Designed pipeline for best-fit parameters of PDS70 water spectral lines

UGA Mathematics Directed Reading: Monoidal Categories: Applications to Abramsky-Coecke Semantics and Deep Learning Athens, GA
(Mentee, DRP Mentor: Gabriel Loos Jan 2026 – Present

- *Tensor Categories* by Gruber
- Gavranović et al. 2024

UGA Mathematics Directed Reading: Intuitionism, Lambda Calculus, and Types Athens, GA
(Mentee, DRP Mentor: Michael Klipper Aug 2025 – Dec 2025

- *Constructivism in Mathematics* by Troelstra and van Dalen

- *Type Theory and Formal Proof* by Geuvers and Nederpelt ch 1-3

UGA Mathematics Directed Reading: Legendrian Curves and Knots Athens, GA
Mentee, DRP Mentor: Eduardo Fernandez Fuertes Sep 2024 – Dec 2024

- Studied Legendrian differential equations and their ties to the Fundamental Group and Homotopies

PUBLICATIONS AND PAPERS

1. (submitted) James Squires, Dr. Nicholas Bork Young, Dr. Nandana Weliweriya. Predicting STEM Classroom Attrition Via Data Mining of Social Dynamics and Academic Metrics. June 2026. ASEE Conference. Charlotte, NC.

PRESENTATIONS AND WORKSHOPS

1. **UGA Math Directed Reading Program (DRP)**, Poultry Science Building, Athens, GA, Nov 2025
 Oral Presentation: *Constructivism and Applications*
2. **CURO Summer Fellows Forum**, Delta Innovation Hub, Athens, GA, Jul 2025
 Oral Presentation: *Improving STEM Education by Predicting Classroom Performance Via Statistical Modeling Trained with Student Social Dynamics*
3. **CURO Symposium Poster Presentation**, Classic Center, Athens, GA, Apr 2025
 Poster Presentation: *Social Dynamics in Introductory Physics Classroom Predictive Analytics*
4. **UGA Math Directed Reading Program (DRP)**, Poultry Science Building, Athens, GA, Dec 2024
 Oral Presentation: *Legendrian Curves and Knots*
5. **CURO Symposium Poster Presentation**, Classic Center, Athens, GA, Apr 2024
 Poster Presentation: *STEM Education Predictive Statistical Learning Model*

HONORS AND AWARDS

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| 1. CURO Research Award (Spring 2026) | December 2025 |
| 2. Barry Goldwater Scholarship —UGA Nominee (2026) | December 2025 |
| 3. CURO 2025 Summer Research Fellow | May 2025 |
| 4. CURO Research Award (Fall 2024) | Aug 2024 |
| 5. Zell Miller Scholar | January 2023 |

TEACHING AND SERVICE

- **Formal Methods and AI Interpretability Directed Reading Group at UGA**
Founder, Director Jan 2026 - Present
- **Physics Directed Reading Program (DRP) at UGA**
Founder, Director Aug 2025 - Present
- **Physics Directed Reading: Mathematical Logic in Physics**
DRP Mentor Jan 2026 – Present
- **Physics Mentor Program (COSMOS)**
Mentor Aug 2024 – Aug 2025

TECHNICAL SKILLS

- Coding Languages: Lean4, C++, Python
- Python Libraries: sklearn, SciPy, ReservoirPy, AstroPy
- Languages: Ancient Greek